

2025 AMC 10B Problem 21 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 21. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 21



Problem:

Each of the 9 squares in a 3×3 grid is to be colored red, blue, or yellow in such a way that each red square shares an edge with at least one blue square, each blue square shares an edge with at least one yellow square, and each yellow square shares an edge with at least one red square. Colorings that can be obtained from one another by rotations and/or reflections are considered the same. How many different colorings are possible?

Answer Choices:

- A. 3
- B. 9
- C. 12
- D. 18
- E. 27

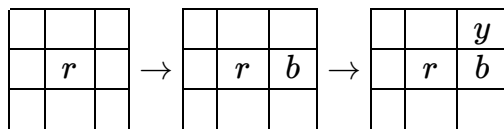
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Stuck on this problem or want to share your approach?

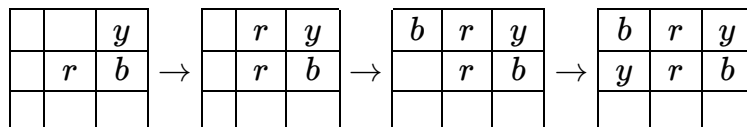
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Solution:

Without loss of generality, assume the center color is red. One of the 4 bordering cells must be blue and one of the two remaining cells bordering the blue must be yellow. Without loss of the generality, suppose the cell to the right of red is blue. Throughout the course of the proof, we will show that none of the other 3 cells bordering the center red can be blue, allowing us to make this designation. Furthermore, suppose there is a yellow above the blue.

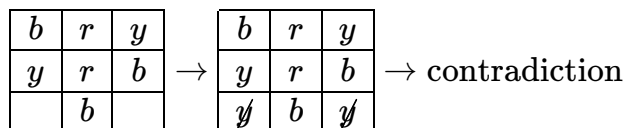


This forces three more moves as the yellow needs to border a red, the new red must border a blue, and the new blue must border a red:



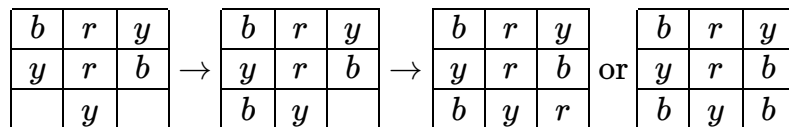
Now we case on the bottom middle square:

Suppose it is blue, then it must neighbor a yellow but placing yellow in either corner would be placing a yellow that doesn't border a red:

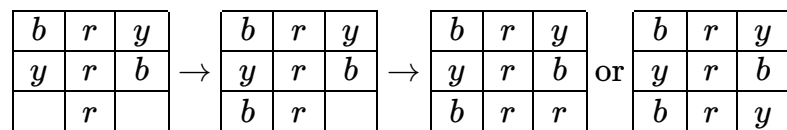


This completes the justification of our WLOG at the beginning.

Suppose the bottom middle square is yellow. The only color that could go in the bottom left is blue and the bottom right square can then be red or blue



Suppose the bottom middle square is red. Then it must border a blue which must border a yellow so the bottom left square must be blue. The bottom right square then can be red or yellow.



So far we have fixed the rotations, but not the reflections and have to consider the scenario where at the beginning there is a yellow under the blue. To deal with this, we would like to flip all solutions that may have y in the bottom right corner so that the y is in the top right corner, except there is a case where there are y 's in both. This case, however, is symmetric up to reflection, so we may take either possibility. Multiplying by 3 for the middle element gives an answer of $4 \cdot 3 = \boxed{\text{(C) } 12}$.

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